Delta Method

$\{X_n\}$ is a sequence of random variables satisfying CLT:

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2)$$

then

$$\sqrt{n}[g(X_n) - g(\theta)] \xrightarrow{d} N(0, \sigma^2 \cdot [g'(\theta)]^2)$$

for any function g satisfying the property that its first derivative is continuous and non-zero valued at θ.

<u>proof</u>

By Taylor's formula we have

$$g(X_n) - g(\theta) = g'(\tilde{\theta})(X_n - \theta)$$

i.e.

$$\sqrt{n}[g(X_n) - g(\theta)] = g'(\tilde{\theta})\sqrt{n}(X_n - \theta)$$

where θ lies between X_n and θ.

$X_n \xrightarrow{p} \theta$, $|\tilde{\theta} - \theta| < |X_n - \theta|$ $g'(\theta)$ continuous yields

$$g'(\tilde{\theta}) \xrightarrow{p} g'(\theta)$$

By Slutsky's theorem, we then have

$$\sqrt{n}[g(X_n) - g(\theta)] \xrightarrow{d} N(0, \sigma^2 \cdot [g'(\theta)]^2)$$